

Quantum 2 HW5

Daniel Haim Breger, 316136944

Technion

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1 The Aharonov-Bohm Effect

1.1 Hamiltonian

Since the magnetic flux is only through the middle of the ring, there is no magnetic field (only a potential) on the ring itself, therefore the hamiltonian won't include an element for the interaction of the spin and the magnetic field. Therefore

$$H = \frac{1}{2m} (\mathbf{p} - e\mathbf{A})^2 \quad (1)$$

inserting the vector potential we have

$$H = \frac{1}{2m} \left(\mathbf{p} - \frac{e\Phi}{L} \right)^2 \quad (2)$$

we can constrain the momentum element because we know the particle is on the ring, therefore

$$\mathbf{p} = -\frac{\hbar}{i} \partial_x = -\frac{\hbar}{iR} \partial_\theta \quad (3)$$

so the hamiltonian is

$$H = \frac{\hbar^2}{2mR^2} \left(-i\partial_\theta + \frac{e\Phi}{2\pi\hbar} \right)^2 \quad (4)$$

and the Schrodinger equation is

$$i\hbar\partial_t\psi = \frac{\hbar^2}{2mR^2} \left(-i\partial_\theta + \frac{e\Phi}{2\pi\hbar} \right)^2 \psi \quad (5)$$

1.2 Setting X

I already did this in the previous subsection. x is defined in $[0, L]$. We can define the basic flux unit $\Phi_0 = 2\pi\hbar/e$ and find

$$H = \frac{\hbar^2}{2mR^2} \left(-i\partial_\theta + \frac{\Phi}{\Phi_0} \right)^2 \quad (6)$$

1.3 Energy Spectrum

We'll perform a gauge transform, defining $\varphi = \frac{\Phi}{\Phi_0}$

$$\psi(\theta) = e^{-i\varphi\theta} \chi(\theta) \quad (7)$$

Which gives us

$$(-i\partial_\theta + \varphi) e^{-i\varphi\theta} \chi(\theta) = (-\varphi\chi(\theta) - i\chi'(\theta) + \varphi\chi(\theta)) e^{-i\varphi\theta} \quad (8)$$

$$= -ie^{-i\varphi\theta} \chi'(\theta) \quad (9)$$

therefore

$$(-i\partial_\theta + \varphi) \psi(\theta) = -ie^{-i\varphi\theta} \partial_\theta \chi(\theta) \quad (10)$$

if we apply the operator again

$$(-i\partial_\theta + \varphi)^2 \psi(\theta) = e^{-i\varphi\theta} (-i\partial_\theta)^2 \chi(\theta) \quad (11)$$

so the Schrodinger equation becomes

$$\frac{\hbar^2}{2mR^2} (-i\partial_\theta)^2 \chi(\theta) = E\chi(\theta) \quad (12)$$

which is the Schrodinger equation for a free particle. This also means χ has a solution of the form

$$\chi(\theta) = e^{ik\theta} \quad (13)$$

Next we consider the boundary conditions which changed due to the gauge transform. We know the problem has periodic boundary conditions, therefore

$$e^{-i\varphi\theta} \chi(\theta) = e^{-i\varphi(\theta+2\pi)} \chi(\theta+2\pi) \quad (14)$$

rearranging

$$\chi(\theta) = e^{-2\pi i\varphi} \chi(\theta+2\pi) \quad (15)$$

inserting the solution ansatz

$$e^{ik\theta} = e^{-2\pi i\varphi} e^{ik(\theta+2\pi)} \implies e^{2\pi i\varphi} = e^{2\pi ik} \quad (16)$$

therefore

$$k = n + \varphi \quad n \in \mathbb{N} \quad (17)$$

and this gives us the energy spectrum

$$E_n = \frac{\hbar^2}{2mR^2} (n + \varphi)^2 \quad (18)$$

as needed.

1.4 Gauge transform

I already showed the energy spectrum in the previous subsection.

1.5 Drawing the Spectrum

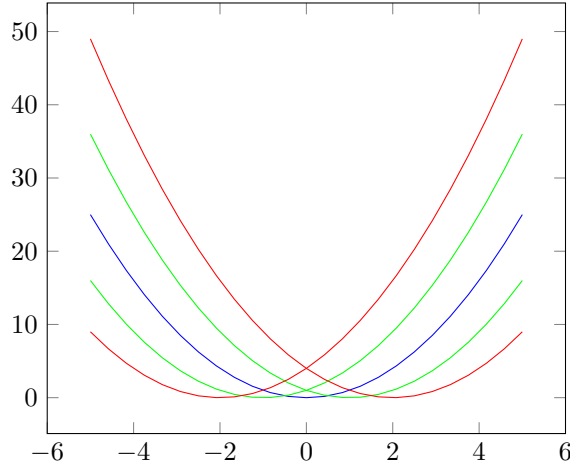


Figure 1 – The energy spectrum described by equation 18 for $n = 0, \pm 1, \pm 2$ marked by blue, green, and red respectively, in multiples of $\frac{\hbar^2}{2mR^2}$. I drew this figure using tikz.

1.6 Reduced Representation

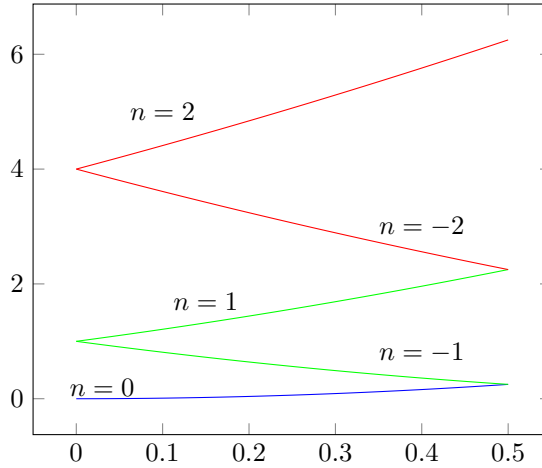


Figure 2 – The energy spectrum described by equation 18 for $n = 0, \pm 1, \pm 2$ marked by blue, green, and red respectively, in multiples of $\frac{\hbar^2}{2mR^2}$, in the domain $0 - 0.5$. I drew this figure using tikz.

The reason we can reduce our focus on just $0 - 0.5$ is that the energy spectrum is periodic in terms of φ . Since it depends on two numbers, one continuous (φ) and one discrete (n), any value of φ over 0.5 is the same as taking a φ less than one half and adding a negative n .

1.7 Odd and Even Energy Spectrum

We'll define

$$n(p) = \begin{cases} \frac{p}{2} & p \text{ is even} \\ -\frac{p+1}{2} & p \text{ is odd} \end{cases} \quad (19)$$

and insert it into equation 18

$$E_p = \frac{\hbar^2}{2mR^2} \begin{cases} \left(\frac{p}{2} + \varphi\right)^2 & p \text{ is even} \\ \left(-\frac{p+1}{2} + \varphi\right)^2 & p \text{ is odd} \end{cases} \quad (20)$$

but we can take out the minus sign from the parentheses and have

$$E_p = \frac{\hbar^2}{2mR^2} \begin{cases} \left(\frac{p}{2} + \varphi\right)^2 & p \text{ is even} \\ \left(\frac{p+1}{2} - \varphi\right)^2 & p \text{ is odd} \end{cases} \quad (21)$$

because $-1^2 = 1$.

1.8 Degeneracy

1.8.1 For $\varphi = 0$

The energy level for $p = 0$ will have a degeneracy of 1, the rest will have a degeneracy of 2.

1.8.2 For $\varphi = \frac{1}{2}$

If we insert $\varphi = \frac{1}{2}$ into the energy levels we find

$$E_p = \frac{\hbar^2}{2mR^2} \begin{cases} \left(\frac{p+1}{2}\right)^2 & p \text{ is even} \\ \left(\frac{p}{2}\right)^2 & p \text{ is odd} \end{cases} \quad (22)$$

which is identical to the case $\varphi = 0$

1.8.3 For $0 < \varphi < \frac{1}{2}$

All energy levels will have a degeneracy of 1, because no two additions of p and φ will match for different p s.

2 Positronium